De Mathematics Competitions

4th Annual

DMC 10 B

Friday, October 23, 2026



INSTRUCTIONS

1. DO NOT TURN TO THE NEXT PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 25-question multiple-choice competition. For each question, only one answer choice is correct.
3. Mark your answer to each problem on the answer sheet with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet will be scored.
4. SCORING: You will receive 6 points for each correct answer, 1.5 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. No calculators, smartwatches, phones, or computing devices are allowed. No problems on the competition will require the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. Before beginning the competition, your competition manager will ask you to record your name on the answer sheet.
8. You will have 75 minutes to complete the competition once your competition manager tells you to begin.
9. When you finish with the competition, sign your name in the space provided on the answer sheet and complete the demographic information questions on the back of the answer sheet.

The problems and solutions for this DMC 10 were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

The DMC Committee reserves the right to disqualify scores from a school if it determines that the rules or the required security procedures were not followed.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

Students who score well on this DMC 10 will be invited to take the De Invitational Mathematics Examination (DIME), which is already out and is due on Friday, January 29, 2027. More details about the DIME can be found on our site.

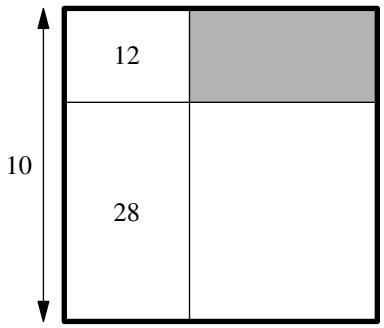
1. A smoothie recipe uses strawberries and bananas in the ratio $5 : 2$. How many strawberries are needed for 10 bananas?

   (A) 4 **(B)** 6 **(C)** 10 **(D)** 15 **(E)** 25

2. Which of the following is always greater than 0 when x and y are real numbers less than 0?

   (A) $\dfrac{2x+y}{x^2}$ **(B)** $x-2y$ **(C)** $\dfrac{y}{x-y^2}$ **(D)** x^2+y **(E)** $xy+x$

3. A square with side length 10 is split into four rectangles by two lines parallel to the sides of the square. Two of those rectangles have areas 12 and 28, as shown below. What is the area of the shaded rectangle?



   (A) 12 **(B)** 15 **(C)** 18 **(D)** 24 **(E)** 30

4. What is the least possible value of

$$|n-2|! + |n-0|! + |n-2|! + |n-6|!$$

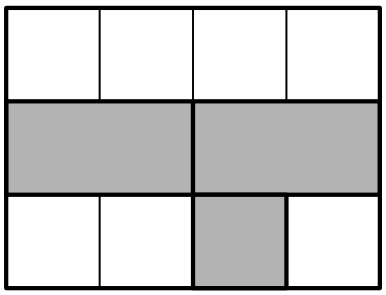
   over all integers n?

   (A) 9 **(B)** 14 **(C)** 28 **(D)** 49 **(E)** 123

5. The 100 floors of a building are labeled, from the bottom up, G, 1, 2, 3, 5, 6, and so on, skipping every number that contains the digit 4. What is the label of the top floor?

   (A) 111 **(B)** 119 **(C)** 120 **(D)** 121 **(E)** 125

6. Three tiles are placed on a 3×4 grid, as shown below in gray. In how many ways can the remaining squares be covered with 1×1 and 1×2 tiles so that no tiles overlap?



(A) 8 **(B)** 9 **(C)** 10 **(D)** 12 **(E)** 15

7. A club has n members who split into 8 teams of equal size. The members within one of the teams play a round-robin tournament, in which every two of them play each other exactly once. Each member of the other seven teams referees exactly one game, and no two members referee the same game. What is the least possible value of n?

(A) 96 **(B)** 104 **(C)** 112 **(D)** 120 **(E)** 128

8. Olivia and Mason each solve math problems at a constant rate. Together they solve 128 problems every 10 minutes, and Mason takes 5 seconds longer per problem than Olivia. How many seconds does Olivia take per problem?

(A) 7.5 **(B)** 10 **(C)** 12.5 **(D)** 15 **(E)** 17.5

9. What is the value of the expression

$$-(-(\cdots-(-(-1+1)+1)\cdots+1)+1),$$

where the digit 1 appears 2026 times, including the 1 in the innermost -1?

(A) -2026 **(B)** -1 **(C)** 0 **(D)** 1 **(E)** 2026

10. Let $P(a, b)$ be a point in the xy-plane above the x-axis and to the right of the y-axis such that when P is reflected over the line $y = x$, its image lies on the line $y = 4x$, and when P is reflected over the x-axis, its image lies on the line $y = -\frac{x}{5} - 1$. What is $a + b$?

(A) 10 **(B)** 15 **(C)** 20 **(D)** 25 **(E)** 30

11. At how many points do the graphs of $y = |x^2 - 4|x||$ and $y = |x|$ intersect?

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 7

12. A triangle has side lengths 4, 5, and 7. For each vertex, Ken chooses to either draw a circle centered there with positive integer radius, or do nothing. No two circles may overlap, though they may be tangent, and no circle may contain any vertex of the triangle other than its center strictly inside it. What is the greatest possible sum of the areas of the circles?

    (A) 25π **(B)** 26π **(C)** 29π **(D)** 34π **(E)** 37π

13. Ten people sit in a row. Each person looks either to the left or to the right and writes how many people are in the direction they are looking. Exactly n people look to the right, and the sum of the ten numbers written is 55. What is the sum of all possible values of n?

    (A) 14 **(B)** 16 **(C)** 18 **(D)** 20 **(E)** 22

14. Infinitely many piles of pebbles sit in a row. The leftmost pile starts with 2026 pebbles, and every other pile is empty. A move is defined as follows: two pebbles are removed from the leftmost pile with at least two pebbles, and one pebble is placed in the pile immediately to its right. Moves are made repeatedly until no moves are possible. How many moves are made?

    (A) 2018 **(B)** 2019 **(C)** 2020 **(D)** 2024 **(E)** 2025

15. How many ordered pairs (m, n) of positive integers are there such that m^2n and mn^2 are both divisors of 216, and m^2n^2 is a divisor of 144?

    (A) 9 **(B)** 12 **(C)** 16 **(D)** 18 **(E)** 24

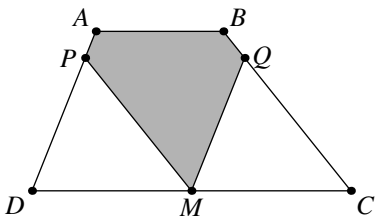
16. For how many positive integers $n \leq 100$ is

$$|1^n - n| + |2^n - 2n| + |4^n - 4n|$$

    one less than a multiple of 5?

    (A) 25 **(B)** 26 **(C)** 30 **(D)** 33 **(E)** 50

17. Trapezoid $ABCD$ has $AB = 2$, $CD = 5$, and $\overline{AB} \parallel \overline{CD}$. Point M is the midpoint of $\overline{CD}$, and points P and Q lie on $\overline{AD}$ and $\overline{BC}$, respectively, so that $\overline{MP} \parallel \overline{BC}$ and $\overline{MQ} \parallel \overline{AD}$, as shown below. What fraction of the trapezoid is shaded?



    (A) $\frac{1}{4}$ **(B)** $\frac{25}{84}$ **(C)** $\frac{1}{3}$ **(D)** $\frac{5}{14}$ **(E)** $\frac{17}{42}$

18. For each positive integer m, let $f(m)$ be the result when the first digit of m is moved to the end. For example, $f(123) = 231$ and $f(2026) = 0262 = 262$. What is the sum of the digits of the greatest four-digit positive integer n for which $n + f(n)$ is a perfect square?

    (A) 17 **(B)** 23 **(C)** 25 **(D)** 27 **(E)** 29

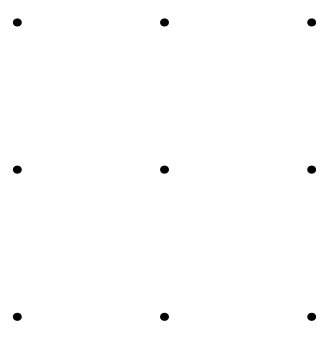
19. There are four creatures in a line: a ghost, a goblin, a scarecrow, and a zombie, where exactly two of the four creatures visited the haunted house. All creatures always tell the truth, and they speak as follows:

    - The ghost: "There is a creature immediately to my left. If that creature visited the haunted house, then I did not visit the haunted house."
    - The goblin: "I did not visit the haunted house."
    - The scarecrow: "If the ghost did not visit the haunted house, then the goblin must have visited the haunted house."
    - The zombie: "There is a creature immediately to my right. If that creature did not visit the haunted house, then I also did not visit the haunted house."

    In how many different orders could the creatures be standing?

    (A) 6 **(B)** 8 **(C)** 9 **(D)** 10 **(E)** 12

20. The nine points below form a 3×3 grid in which neighboring points are 1 unit apart. Segments are drawn between pairs of these points, where each segment must have length 1 or $\sqrt{2}$, and no segment is drawn more than once. What is the least positive integer n such that whenever n such segments are drawn, some three of them are the sides of a triangle?

    (A) 13 **(B)** 14 **(C)** 15 **(D)** 16 **(E)** 17

21. For how many integers a does the equation

$$x^4 + ax^3 + 2026x^2 + ax + 1 = 0$$

    have four distinct real solutions?

    (A) 1846 **(B)** 1848 **(C)** 1850 **(D)** 1852 **(E)** infinitely many

22. Daniel cuts out 9 paper shapes, each of area 3, and places them flat on a table. Some of the shapes overlap, and the total area of the table covered by the shapes is 22. Let A be the greatest real number with the property that, no matter how the shapes are placed, there exist some two shapes whose intersection has area at least A. Then A can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m+n$?

(A) 14 **(B)** 23 **(C)** 32 **(D)** 41 **(E)** 50

23. In rectangle $ABCD$ with area 8, let M be the midpoint of $\overline{AB}$. There exists a point E on $\overline{AD}$ such that $\overline{BE} \perp \overline{CM}$ and $\overline{CE}$ bisects $\angle BED$. What is the area of $\triangle CME$?

(A) $2\sqrt{2}$ **(B)** 3 **(C)** $2\sqrt{3}$ **(D)** 4 **(E)** $3\sqrt{2}$

24. A machine has one button labeled $2^a \cdot 3^b$ for each pair of integers $a \geq 0$ and $b \geq 1$. Each minute, the machine shuts off permanently with probability $\frac{1}{21}$. Otherwise, Taiki presses exactly one button, choosing the button labeled $2^a \cdot 3^b$ with probability $2^{-a} \cdot 3^{-b}$. What is the probability that, before the machine shuts off, Taiki has pressed at least three (not necessarily distinct) buttons whose labels are divisible by 36?

(A) $\frac{125}{2744}$ **(B)** $\frac{27}{512}$ **(C)** $\frac{1}{8}$ **(D)** $\frac{125}{512}$ **(E)** $\frac{343}{1331}$

25. Ryan writes each of the 12 positive integer divisors of 72 on a whiteboard. Each round, he chooses one of the numbers remaining on the board uniformly at random and erases it along with all of its multiples. The expected number of rounds until the board is empty is $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m+n$?

(A) 346 **(B)** 347 **(C)** 348 **(D)** 349 **(E)** 350